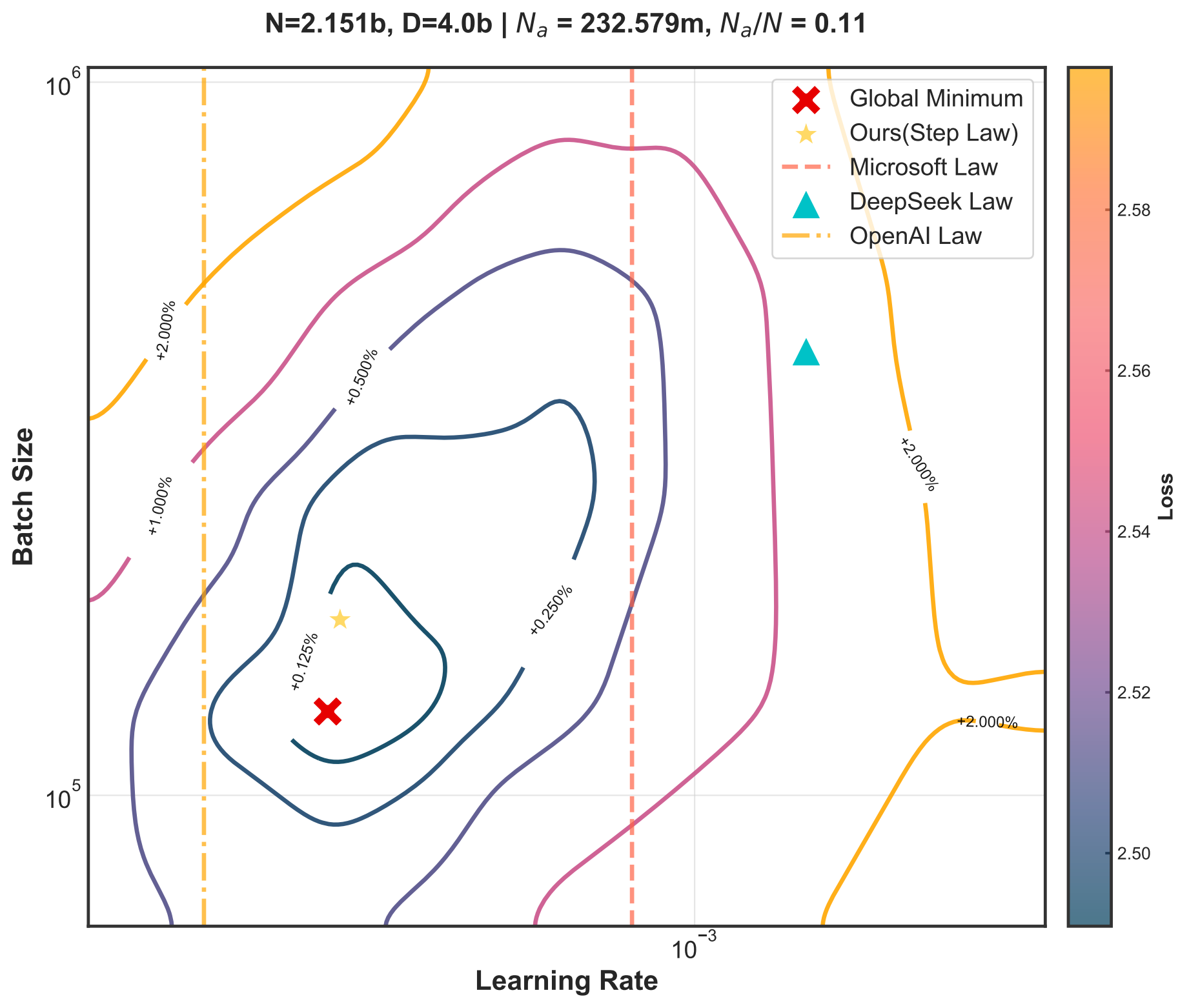
N=2.151b, D=4.0b | N_a = 232.579m, N_a/N = 0.11
Global Minimum
Ours(Step Law)
Microsoft Law
DeepSeek Law
OpenAI Law
+2.000%
+1.000%
+0.500%
+0.250%
+0.125%
+2.000%
+2.000%
Batch Size
10^6
10^5
10^{-3}
Learning Rate
Loss
2.58
2.56
2.54
2.52
2.50